

Predictive Analytics Beyond the Hype: A Comprehensive Comparison of LSTM, XGBoost and LightGBM with Emphasis on RMSE and CPU Utilization

Abhinav Chola

Central Research Laboratory Bharat Electronics Ltd. Ghaziabad
abhinavchola@bel.co.in

Rupali Rastogi

Central Research Laboratory Bharat Electronics Ltd. Ghaziabad
rupalirastogi@bel.co.in

Pushpinder Kaur

Central Research Laboratory Bharat Electronics Ltd. Ghaziabad
pushpinderkaur@bel.co.in

Aman Chaudhary

Central Research Laboratory Bharat Electronics Ltd. Ghaziabad
amanchaudhary@bel.co.in

Debasis Biswas

Central Research Laboratory Bharat Electronics Ltd. Ghaziabad
debasisbiswas@bel.co.in

Abstract—Predictive maintenance (PdM) technique leverages data analytics to identify and predict equipment failures. Using data collected from IoT devices such as sensors, PdM applies machine learning techniques, and real-time equipment monitoring to determine exactly when it's the best time to perform equipment maintenance. Anticipating equipment maintenance time beforehand aims to minimize maintenance costs and maximize uptime. Predictive maintenance also aims to predict future adverse events, in order to improvise and plan maintenance schedules in advance. The PdM results are therefore beneficial in reducing downtime and for timely arrangement of essential repair equipment. The maintenance industry was earlier dominated by Preventive and Corrective maintenance practices. The former technique relies on fixed schedule of pre-planned maintenance while the latter requires taking action once a fault occurs. However, in recent times, the increased use of on-board sensors and data-driven algorithms has stimulated the shift to data-driven predictive maintenance approach for aircraft. Leveraging the sensor data, data analytic techniques and Deep Learning models, predictive maintenance approach can calculate the Remaining Useful Lifecycle (RUL) of aircraft engines and can use it to schedule maintenance dynamically.

This paper focuses on analyzing the CMAPSS dataset generated by NASA for better understanding of impact created by sensor data. The aim of this paper is to give deeper insight of sensor data, data pre-processing techniques, data featuring, RUL prediction mechanism for maintenance scheduling. The paper presents a comparative analysis of regression algorithms like LSTM, XGBoost and LightGBM. The evaluation criterion is root mean squared error (RMSE) using Regression Analysis and the corresponding CPU utilization. After comparison it is concluded that LSTM outperforms the other deep learning methods for CMAPSS Sensor data.

Index Terms—Predictive Maintenance, CMAPSS, LSTM, XGBoost, LightGBM, Aircrafts, RUL

I. INTRODUCTION

The day-to-day life is driven by industrial automation and relies heavily on a vast range of mechanical and electrical equipment including agriculture, automobiles, power systems, gadgets etc[1]. In this machine dependent ecosystem, it is

imperative to perform maintenance operation from time-to-time [2]. There exist three primary approaches for undertaking equipment maintenance in industries namely, reactive, preventive and predictive maintenance. In reactive maintenance, repair-related tasks are considered only after certain machine failure has occurred while with preventive maintenance a fixed time-interval based maintenance is performed. The former technique fails to prevent industry accidents while the latter approach costs heavily. Therefore, the ideal way to deal with maintenance is by using predictive maintenance (PdM), a method that provides a pre-emptive estimate of the time-to-failure of an equipment regarding which scheduling maintenance is necessary[3].

This approach emphasizes on predicting the failure time by taking input from various sensors placed on the equipment and analyse it to model degradation trends present between input sensors and time-to-failure duration of the machine.

The primary advantage of relying on prediction-based maintenance strategy is the elimination of unplanned machine downtime, reduction of overall maintenance costs and ultimately maximizing the machine lifetime. All machine dependent domains are now turning towards predictive maintenance and the aviation business is an example of one such industry. Aircraft engine is complex equipment and needs continuous monitoring to ensure good engine performance. In the last decade, the fault diagnostics and prognostics of components related to aircraft engine has gained great attention as it provides a way for huge cost-cuttings [4], [5], [6]. Furthermore, in the last decade, with great strides in deep learning, the old corrective and preventive maintenance has given way to the more sophisticated and dynamic method of equipment maintenance through Predictive maintenance especially regarding aircraft engines.

According to Airbus [7], by year 2025, the number of unscheduled aircraft grounding occurring due to fault repairs may become zero as a result of superior data analytics and PdM based scheduling. Monitoring health of Aircrafts and predicting the ideal time for their maintenance could directly

lead to the elimination of unscheduled groundings of aircrafts and the associated operational interruptions [7,8]. The paper is organized in different sections starting with introduction in section 1. Afterwards, Section 2 presents the detailed literature review on the existing methods useful for turbofan engine RUL estimation. Section 3 explains the dataset taken for experimental testing. Section 4 compiles the proposed methodology with a comprehensive block diagram while Section 5 explains the parameters obtained after optimization. Section 6 and 7 discusses and compares the experimental results and provide a brief conclusion respectively.

II. LITERATURE SURVEY

Predictive Maintenance concerning the aviation industry works on a measuring unit known as Remaining Useful Lifecycle (RUL) and it represents the remaining health of aircraft equipment, especially related to engines. Researches in the PdM field are keen on enhancing the accuracy or minimizing the error with which they can predict RUL values for aircraft equipment. The growth of deep learning technology in recent years has resulted in a dominance of models like LSTM and hybrid-CNN in RUL-based prediction algorithms. This section discusses the various prominent machine learning based algorithms proposed in Predictive Maintenance domain.

Andreas et.al presented a deep learning-based model in which pre-processed data is fed into an effective LSTM network [1]. The LSTM model is further coupled with certain performance booster techniques including piece-wise linear RUL function and dimensionality reduction algorithms to achieve high performance on the standard sensor-based dataset for aircraft turbofan engine. The model is validated on various combinations of hyper parameters and performs in the range of 8-40 RMSSE score. Zhang et al. presented a multi-functioning algorithm to revolutionize the popular deep belief network into numerous networks working in parallel simultaneously to provide results in terms of two goals, accuracy and diversity[4]. Furthermore, the model is robust to noise in input data and is able to manage multiple complicated operations without hampering its performance. Babu et. al, implemented a deep CNN-based regression model that estimates RUL for aircraft components[5].

The network operates on convolutional layers which are two dimensional and are key for extracting important features. The proposed model is also capable of extracting spatial features efficiently. Li et.al, worked on a deep convolutional neural network (CNN) to estimate RUL values[6]. The proposed architecture of CNN is designed to extract features from pre-prepared 2-D sensor data when passed into convolution layers.

David and Skaf proposed a balanced approach to harness a data-dependent deep learning model to correctly predict the right time for replacement of aircraft components especially while using imbalanced data. The presented model outperforms various prominent LSTM- based solutions because of its high efficiency in learning dependencies (temporal) and is able to manage volatility dynamics features while using time-series datasets[7].

The model when used with datasets with an imbalanced ratio of as low as 5% performed better than certain state of the art methods and predicted aircraft component health with 70% test accuracy. Sam, Jaison et.al proposed an end-to-end approach leveraging two aircraft maintenance data sets providing historical maintenance data, processed with multiple analytical methods and a digital twin to provide a comprehensive outlook at the real-time aircraft health to end user[8]. The experimental setup uses both datasets as input for a neural network using a software called CORTEX which unify machine learning along with data visualization to provide real-time predictive, and actionable insights regarding aircraft maintenance plans. The accuracy of the proposed technique when tested over various aircraft components is on average around 88

III. CMAPSS DATASET FOR TURBOFAN ENGINE

A consistent research gap noticeable in most prior works in this field is the lack of actual turbofan engine diagram and the meaning of sensors involved in CMAPSS dataset[9]. This section aims to offer a comprehensive analysis of the turbofan engine along with its components, sensors and their functioning.

In 2008, NASA generated the C-MAPSS dataset in MATLAB environment by simulating real-time turbofan engines. The name C-MAPSS stand for Commercial Modular Aero-Propulsion System Simulation and is used as the benchmark dataset for all aircraft based predictive maintenance technique development and analysis. This paper utilizes FD001 dataset which is a sub-set of the popular CMAPSS dataset. FD001 contains historic information regarding 100 Engine Units each of which have 21 sensors, a single sea-level operating condition and a single HPC Degradation fault mode, all present in time series format. Each time series represents a different engine therefore signifying that the data can be understood to of a fleet of similar engines. Each row present in the dataset showcases unit number of the engine, Time cycles, three unique operational settings and readings from 21 sensors. The last column of all rows represents Remaining Useful Lifecycle (RUL) value of that engine for corresponding sensor values of the engine. The RUL value of an equipment (here engine) represents the possible number of flights, after the last data point in the test dataset, that engine can support without failing. The standard turbofan engine comprises of multiple key components such as fan, low pressure turbine, low, high pressure turbine, high pressure compressor, etc. The positioning of individual component and the twenty one sensors in the turbofan engine are shown in "Fig. 4".

A low pressure compressor (LPC) and high pressure compressor (HPC) are responsible for providing gases at compressed high temperature and high pressure directly to the combustor. Moreover, the Low pressure turbine (LPT) works to decelerate the air flow and pressurize it to enhance the process of converting chemical energy of aviation fuel(kerosene). The High pressure turbines (HPT) on the other hand generate strong mechanical energy by striking gas at high pressure and temperature to turbine blades. Meanwhile, Low-pressure rotor

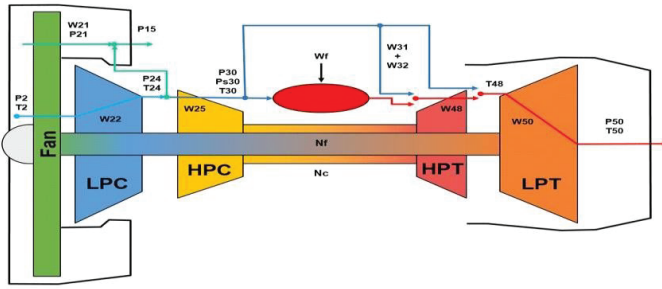


Fig. 1. Diagram showing parts of Turbofan engine

(N1), high-pressure rotor (N2), and nozzle ensure that engine performs at good combustion efficiency. The various sensors comprising a turbofan engine are presented in Table 1.

TABLE I
TABLE TYPE STYLES

S. No	Sensor Details		
	Sensor Name	Symbol	Description
1	Sensor 1	T2	Fan inlet temp
2	Sensor 2	T24	LPC Outlet temp
3	Sensor 3	T30	HPC outlet temp
4	Sensor 4	T50	LPT outlet temp
5	Sensor 5	P2	Fan inlet pressure
6	Sensor 6	P15	Bypass Duct Ratio
7	Sensor 7	P30	HPC outlet pressure
8	Sensor 8	Nf	Physical Fan Speed
9	Sensor 9	Nc	Physical Core Speed
10	Sensor 10	EPR	Engine Pressure Ratio
11	Sensor 11	Ps30	HPC outlet static pressure
12	Sensor 12	Phi	Ratio of fuel flow to Ps30
13	Sensor 13	NRf	Corrected Fan Speed
14	Sensor 14	NRc	Corrected Core Speed
15	Sensor 15	BPR	Bypass Ratio
16	Sensor 16	farB	Burner Fuel Air Ratio
17	Sensor 17	HtBleed	Bleed Enthalpy
18	Sensor 18	Nf_dmd	Demanded Fan Speed
19	Sensor 19	PCNfR_dmd	Demanded Corrected Fan Speed
20	Sensor 20	W31	HPT Coolant Bleed
21	Sensor 21	W32	LPT Coolant Bleed

IV. PROPOSED METHODOLOGY

Traditional approach for predictive maintenance relied on machine learning techniques in which models required manual feature engineering. This involved manual construction of adequate features which requires vast domain expertise. Moreover, such models were not prone to reusability since feature engineering is highly specific to individual problems and data availability, both of which vary for different end users. Therefore researches have turned to Deep learning models for deriving PdM solutions.

In this paper, a unique comparison of three deep learning models namely LSTM, XgBoost, and LightGBM is presented on CMAPSS dataset[10]. Such comparison is not found in the existing literature in the field of PdM of aviation. Furthermore, gradient search is implemented for all the three models and the best parameter values discovered after hyper-parameter tuning is utilized to predict RUL values[11].

A. Deep Learning

The following three deep learning models are used in this paper to analyse comparative performance in the field of PdM in aeronautics.

1) *LSTM*: Long Short Term Memory (LSTM) network works as a special form of recurrent neural network with the potential of learning both short term and long term dependencies[12]. This implies, LSTM models can find usage in various domains including machine failure records, monitoring electricity consumption, predicting stock market prices, and so on.

Since applications built on time-series data can benefit by deploying LSTM model to look back into longer time periods when finding patterns to detect equipment failure. Therefore, Long Short Term Memory LSTM networks with their ability to learn from sequential data are tailor-made for researchers working in the predictive maintenance vertical.

In this paper, a five layered LSTM model is deployed to predict RUL of aircraft components using CMAPSS dataset. The LSTM architecture as shown in fig. comprises of two pairs of LSTM and dropout layers converging into a fifth dense layer which finally provides the RUL output. The data from FD001_train is preprocessed using extensive correlation analysis and PCA. As a result, seven columns of FD001_train are dropped and the remaining fourteen sensor data values are used for LSTM model training and testing.

2) *XGBoost*: XGBoost is supervised gradient boosting technique used widely for machine learning and data science applications. It essentially provides an efficient, robust and faster implementation of gradient boosting mechanism[13]. Ensemble learning provides a comprehensive solution by unifying the prediction powers of multiple models and mould it into a single model eventually. The obtained single model offers the aggregated output of numerous models which if used individually could only provide results a little better than random guessing. XGBoost is a treebased algorithm and functions as an ensemble model. This implies, XGBoost is a combination of predictors working towards a unified objective of accurately predicting a common target variable. The various predictions are aggregated and one final prediction is delivered as an output of the model. XGBoost involves training a varying number of decision trees such that each tree is to be trained on a subset of the data. The final prediction is generated by merging the predictions from each decision tree. Since it operates on ensemble learning, XGBoost has proven to perform better than popular machine learning models in a both classification and regression related scenarios.

3) *Light GBM*: Light GBM is popular less-complex framework for a gradient-boosting algorithm that operates on tree based learning mechanism[14]. Unlike other tree-based algorithms which develop trees horizontally, Light GBM involves vertical tree growth. This implies Light GBM follows a leaf-wise tree growth approach rather than the traditional level-wise and chooses the leaf with max delta loss to grow. The difference in approach is critical as an algorithm dependent on leaf-wise growth can reduce more loss as compared to a

level-wise growth algorithm. A few important parameters and their usage is listed below:

- **max_depth**: It acts as a limiter for the treedepth and has a default value of 20. Lower values of max_depth can help reduce the risk of over fitting.
- **categorical_feature**: It emphasizes on the primary categorical feature to be used for training the model.
- **bagging_fraction**: It defines the percentage of data to be used for individual iteration.
- **num_iterations**: It defines the amount of iterations to be performed by the model during training. The default value is 100 which can be increased if early- stopping is involved.
- **num_leaves**: It limits the quantity of leaves to be present in a tree. This parameter is allowed to have a value less than the square of max_depth parameter.
- **max_bin**: It defines the maximum bins to bucket the used feature values.
- **min_data_in_bin**: It provides the minimum quantity of data allowed in one bin.
- **task**: It is set to define the task chosen between training and prediction for the model in use. This parameter uses train as its default setting.
- **feature_fraction**: It specifies the fraction of features to be considered in each iteration. The default value is one.

B. Data Pre-Processing Techniques

Dimensionality reduction is a popular method used for decreasing the available feature quantity in a dataset while retaining major parts of key information. Essentially, this technique inputs high-dimensional data, transforms it and outputs a lower-dimensional data that still carries the key essence of original data. Since, the available sensor data is vast and only parts of it play a key role in generating accurate predictions, dimensionality reduction technique is used to filter out and discard the sensor input that has minimum contribution to RUL estimation[15]. Another popular technique for data-preprocessing is feature selection. It is an efficient process to filter relevant features of the original dataset by ejecting noisy and irrelevant data. The goal of feature selection is to traverse the complete dataset and obtain the minimum number of features that can bring maximum accuracy for machine learning models. This technique identifies a feature subset that could comprehensively depict all the relevant information from the dataset. Feature selection is heavily used in applications related to data mining, machine learning, artificial intelligence, etc. The dimensionality reduction process is followed by feature selection via correlation and therefore forms the following two steps:

- Step 1: Standard deviation is measured and sensors with minimum fluctuations in standard deviation are discarded to achieve dimensionality reduction[16].
- Step 2 is carried for sensors remaining after the above step, correlation analysis is performed to further reduce the dimensionality. The sensors possessing a low corre-

lation with RUL are dropped and rests of the features are selected.

After performing step 1, it is concluded that standard deviation of columns namely settings 1 and settings 2, as shown in fig, is negligible so both features operational settings 1 and 2 can be discarded from the dataset. Moreover, standard deviation values of sensors 1,5, 10,16,18,19 offer none or minimum fluctuation, so they can also be discarded to achieve dimensionality reduction.

	count	mean	std	min	25%	50%	75%	max
s_1	20631.0	518.670000	0.000000e+00	518.6700	518.6700	518.6700	518.6700	518.6700
s_2	20631.0	642.680934	5.000533e-01	641.2100	642.3250	642.6400	643.0000	644.5300
s_3	20631.0	1590.523119	6.131150e+00	1571.0400	1586.2600	1590.1000	1594.3800	1616.9100
s_4	20631.0	1408.933782	9.000605e+00	1382.2500	1402.3600	1408.0400	1414.5550	1441.4900
s_5	20631.0	14.620000	1.776400e-15	14.6200	14.6200	14.6200	14.6200	14.6200
s_6	20631.0	21.609803	1.388985e-03	21.6000	21.6100	21.6100	21.6100	21.6100
s_7	20631.0	553.367711	8.850923e-01	549.8500	552.8100	553.4400	554.0100	556.0600
s_8	20631.0	2388.096652	7.098548e-02	2387.9000	2388.0500	2388.0900	2388.1400	2388.5600
s_9	20631.0	9065.242941	2.208288e+01	9021.7300	9053.1000	9060.6600	9069.4200	9244.5900
s_10	20631.0	1.300000	0.000000e+00	1.3000	1.3000	1.3000	1.3000	1.3000
s_11	20631.0	47.541168	2.670874e-01	46.8500	47.3500	47.5100	47.7000	48.5300
s_12	20631.0	521.413470	7.375534e-01	518.6900	520.9600	521.4800	521.9500	523.3800
s_13	20631.0	2388.096152	7.191892e-02	2387.8800	2388.0400	2388.0900	2388.1400	2388.5600
s_14	20631.0	8143.752722	1.907618e+01	8099.9400	8133.2450	8140.5400	8148.3100	8293.7200
s_15	20631.0	8.442146	3.750504e-02	8.3249	8.4149	8.4389	8.4656	8.5848
s_16	20631.0	0.030000	1.387812e-17	0.0300	0.0300	0.0300	0.0300	0.0300
s_17	20631.0	393.210654	1.548763e+00	388.0000	392.0000	393.0000	394.0000	400.0000
s_18	20631.0	2388.000000	0.000000e+00	2388.0000	2388.0000	2388.0000	2388.0000	2388.0000
s_19	20631.0	100.000000	0.000000e+00	100.0000	100.0000	100.0000	100.0000	100.0000
s_20	20631.0	38.816271	1.807464e-01	38.1400	38.7000	38.8300	38.9500	39.4300
s_21	20631.0	23.289705	1.082509e-01	22.8942	23.2218	23.2979	23.3668	23.6184

Fig. 2. Snapshot of datasets during pre-processing

Further, correlation analysis of remaining sensors is shown in Fig 2. The analysis proves that sensor 9 and 14 show maximum correlation, therefore sensor 14 can be dropped (or not as per requirement). After following both dimensionality reduction and feature selection, the 23 features (21 sensors and 2 operations columns) are reduced to the following 14 sensors, as shown in the figure.

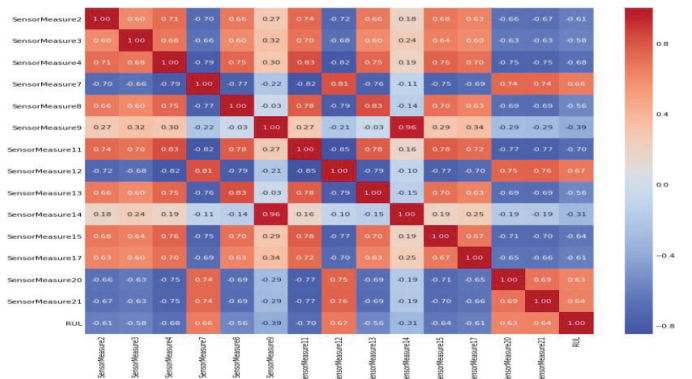


Fig. 3. Snapshot of PCA of dataset

The rightmost label in the Fig 3 represents Pearson correlation coefficient. After the PCA, Sensor 2, Sensor 3, Sensor 4, Sensor 7, Sensor 8, Sensor 9, Sensor 11, Sensor 12, Sensor 13, Sensor 14, Sensor 15, Sensor 17, Sensor 20, Sensor 21 are selected for model input. The above two steps will enhance both accuracy and learning speed of the model.

C. Flowchart of Methodology

All three models presented in the paper broadly follow the same process as shown in the flow diagram in Fig 4. The data from CMAPSS dataset is pre-processed using dimensionality reduction and feature selection, as explained in the previous subsection. Afterwards, the data is split using the standard test-train split function and training data is used for enhancing the model's RUL prediction ability in a supervised form. The trained model is subjected to testing and validation using the test data. The RUL predicted is compared with the actual RUL present in test data for validation.

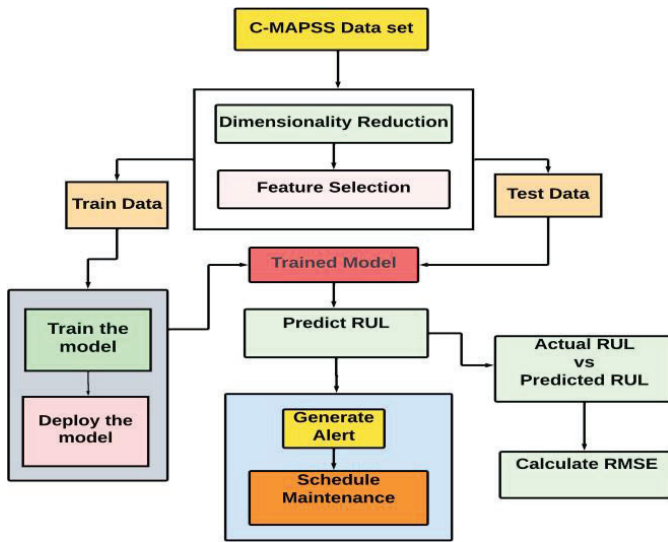


Fig. 4. Flowchart showcasing RUL prediction algorithm for PdM

D. Evaluation Matrix

Root Mean Square Error RMSE is used as the primary validation metric [17]. The RMSE metric operates on the following formula:

$$RMSE = \sqrt{\frac{\sum_{i=1}^n (x_i - y_i)^2}{n}} \quad (1)$$

In the above formula, x and y represent the predicted and actual RUL values respectively and n stands for the number of predictions. Further, based on requirement, the RUL prediction can be used to schedule maintenance or design an alarm system for aircraft companies.

V. MODEL PARAMETERS AND OPTIMIZATION

The pre-processing and model flow described in section IV were followed to implement LSTM, XGBoost and Light

GBM regression models on the CMAPSS dataset (FD001). The models were optimized in different ways and the best case RMSE values were compared.

A. LSTM

A multi-layered LSTM model along with multiple dropout layers to ensure efficient regularisation was used to calculate RUL on given dataset. The model was generated in the following sequence performed in the following sequential combination and with early stopping of patience=20:

- Layer 1: LSTM layer with units=100 and return sequence=True.
- Layer 2: Dropout layer with value=0.2
- Layer 3: LSTM layer with units=100 and return sequence=False.
- Layer 4: Dropout layer with value=0.2
- Layer 5: Dense layer with and activation=relu and units=1
- Optimizer : rmsprop

B. XGBoost

XGBoost works on gradient boosting which is a form of supervised learning algorithm that leverages multiple parameters while training and testing the model. The accuracy of XGBoost model depends heavily on the values of these parameters. Therefore, a hyper parameter grid search was performed to tune the required parameters so as to obtain maximum efficiency during RUL prediction by regression mechanism. The final parameters used with the XGBRegressor model had the values as follows, max_depth = 3, learning_rate = 0.1, min_child_weight = 1, subsample = 0.4, eta = 0.01, n_estimators = 1000, gamma = 100, colsample_bytree = 0.2, booster = gbmtree

C. LightGBM

LightGBM model also works on gradient boosting but is different to XGBoost in terms of implementation. Therefore a different set of parameters are required while implementing the model. The parameters of LightGBM were also optimized to prevent overfitting and the following set of parameters were used Boosting_type = gbdt, Metric =11,12, Learning_rate = 0.05, Feature_fraction = 0.6, Bagging_fraction = 0.2, Bagging_frequency = 10, Max_depth =15, Num_leaves = 20, Max_bin = 512, Num_iteration = 100. Along with the parameters, an early stopping of 10 rounds was implemented.

VI. PERFORMANCE EVALUATION OF DEEP LEARNING MODELS

The models described in section IV used the optimized parameters explained in section V. The models were trained and tested on FD001 dataset and were programmed to predict the RUL values of aircrafts and RMSE, CPU Utilization and Memory usage were considered as the performance metrics. Table 2 and Table 3 provide a comparative analysis of resulting output performances.

TABLE II

COMPARING PDM MODELS FOR RUL PREDICTION USING RMSE

S. no	Parameters		
	Model Name	Training RMSE	Testing RMSE
1	LSTM	20.3	33.57
2	XGBoost	36.5	41.3
3	LightGBM	39.02	42

TABLE III

COMPARING PDM MODELS FOR CPU AND MEMORY UTILISATION

S. no	Parameters		
	Model Name	CPU Utilisation (%)	Memory Utilisation (mb)
1	LSTM	5	685
2	XGBoost	900	587
3	LightGBM	17	400

The RMSE values of Table 2 indicates that LSTM with RMSE value of 33.5 performs superior as compared to XGBoost and LightGBM both of which offer a RMSE of approximately 40. However, it is important to note that XGBoost and LightGBM perform at a much higher speed than LSTM and if the dataset is larger, then they have the potential to outperform LSTM. To inspect the model performances further, Table 3 showcases the individual CPU utilization and memory cost of all 3 PDM models. Table 3 supports LSTM and LightGBM as their CPU utilization is much diminished as compared to the 90% consumption of XGBoost. Therefore, after comparing the 3 models on different parameters, it can be concluded that LSTM performs the best for time-series data and is the best suited for aircraft RUL prediction.

VII. CONCLUSION

Equipment health is a major cornerstone in machine dependent businesses such as the aircraft industry. This has prompted companies to move away from the traditional methods and adopt the new-age data driven techniques of equipment health prognostics. Therefore thorough inspection and maintenance of machines plays a significant role in keeping equipment health intact. Predictive Maintenance is one of the most popular methods to schedule cost effective maintenance in the current scenario. This paper discussed the latest method of Predictive maintenance and provided a comparative analysis of three popular deep learning models available in this domain, namely LSTM, XGBoost and LightGBM. The paper provided a brief literature review of the latest work done in Predictive Maintenance (PdM) and further explained the CMAPSS dataset. There is a lack of actual turbofan engine diagram and the actual functioning of sensors involved in CMAPSS dataset, which the proposed paper aims to overcome. Furthermore, the paper provided a comprehensive flowchart that explains the workflow of deep learning models for Predictive Maintenance.

The paper also explains the optimization techniques used for each model and compared the overall results in terms of RMSE and CPU utilization. The comparison proved that LSTM is superior to both XGBoost and LightGBM if CPU based system

is used. However in case of GPU, XGBoost can provide faster results. In future the comparison provided here could be used to select appropriate models to build a hybrid deep learning model that may outperform the current PdM algorithms. Furthermore, the proposed paper can be extended to generate elaborated scheduling algorithms to create a maintenance plan for aircraft fleets.

REFERENCES

- [1] Predictive maintenance enabled by machine learning: Use cases and challenges in the automotive industry Andreas Theissler, Judith Pérez-Velázquez b,c,,Marcel Kettelgerdes b , Gordon Elger b,d,ELSEVIER, 2021.
- [2] F A Adryan,* , K W Sastra,"Predictive Maintenance for Aircraft Engine Using Machine Learning: Trends and Challenges", International Journal of Aviation Science and Engineering, 2021.
- [3] Ehlers, Ruediger. "Formal verification of piece-wise linear feed-forward neural networks." Automated Technology for Verification and Analysis: 15th International Symposium, ATVA 2017, Pune, India, October 3–6, 2017, Proceedings 15. Springer International Publishing, 2017.
- [4] C. Zhang, P. Lim, A. K. Qin and K. C. Tan, "Multiobjective deep belief networks ensemble for remaining useful life estimation in prognostics", IEEE Trans. Neural Netw. Learn. Syst., vol. 28, no. 10, pp. 2306-2318, Oct. 2017.
- [5] J. Zhang, P. Wang, R. Yan and R. X. Gao, "Long short-term memory for machine remaining life prediction", J. Manuf. Syst., vol. 48, pp. 78-86, Jul. 2018.
- [6] A. Elsheikh, S. Yacout and M.-S. Ouali, "Bidirectional handshaking LSTM for remaining useful life prediction", Neurocomputing, vol. 323, pp. 148-156, Jan. 2019.
- [7] X. Li, Q. Ding and J.-Q. Sun, "Remaining useful life estimation in prognostics using deep convolution neural networks", Rel. Eng. Syst. Saf., vol. 172, pp. 1-11, Apr. 2018.
- [8] M. David Dangut, Z. Skaf and I. Jennions, "Rescaled-LSTM for Predicting Aircraft Component Replacement Under Imbalanced Dataset Constraint," 2020 Advances in Science and Engineering Technology International Conferences (ASET), Dubai, United Arab Emirates, 2020.
- [9] Saxena, K. Goebel, D. Simon, and N. Eklund, Damage Propagation Modeling for Aircraft Engine Run-to-Failure Simulation, in the Proceedings of the 1st International Conference on Prognostics and Health Management (PHM08), Denver CO, Oct 2008.
- [10] Heim, S., Clemens, J., Steck, J. E., Basic, C., Timmons, D., Zwiener, K. (2020). Predictive Maintenance on Aircraft and Applications with Digital Twin. 2020 IEEE International Conference on Big Data (Big Data).
- [11] OWAIS ASIF , SAJJAD ALI HAIDER , SYED RAMEEZ NAQVI , JOHN F. W. ZAKI , KYUNG-SUP KWAK , AND S. M. RIAZUL ISLAM,"A Deep Learning Model for Remaining Useful Life Prediction of Aircraft Turbofan Engine on C-MAPSS Dataset", IEEE Access, August 2022.
- [12] Wu, Haiyue, Aihua Huang, and John W. Sutherland. "Layer-wise relevance propagation for interpreting LSTM-RNN decisions in predictive maintenance." The International Journal of Advanced Manufacturing Technology (2022).
- [13] Paliari, Iliana, Aikaterini Karanikola, and Sotiris Kotsiantis. "A comparison of the optimized LSTM, XGBOOST and ARIMA in Time Series forecasting." 2021 12th International Conference on Information, Intelligence, Systems & Applications (IISA). IEEE, 2021.
- [14] Lijun Liu1, Lan Wang, Zhen Yu1 Remaining Useful Life Estimation of Aircraft Engines Based on Deep Convolution Neural Network and LightGBM Combination Model, International Journal of Computational Intelligence Systems, 2021.
- [15] Deng, Kunyuan, et al. "A remaining useful life prediction method with long-short term feature processing for aircraft engines." Applied Soft Computing 93 (2020).
- [16] Chang Woo Hong, Changmin Lee, Kwangsuk Lee, Min-Seung Ko, Dae Eun Kim, and Kyeon Hur " Remaining Useful Life Prognosis for Turbofan Engine Using Explainable Deep Neural Networks with Dimensionality Reduction", MDPI, 2020.
- [17] Chai, Tianfeng, and Roland R. Draxler. "Root mean square error (RMSE) or mean absolute error (MAE)." Geoscientific model development discussions 7.1 (2014).